

Factuality-Aware Reranking for Extreme Summarization

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Problem and Setup

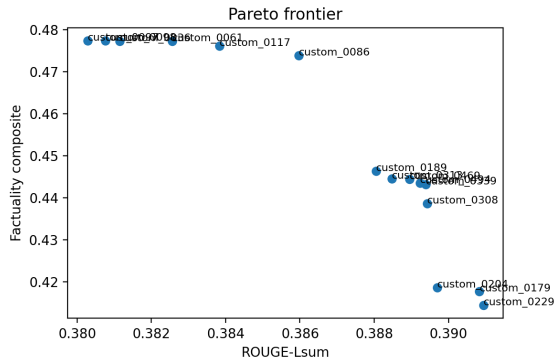
- ▶ XSum rewards compression, so fluent hallucinations are common.
- ▶ Overlap metrics alone do not tell us whether a summary is supported by the article.
- ▶ Keep the public facebook/bart-large-xsum path as the baseline.
- ▶ Ask whether reranking can trade a small amount of ROUGE for better factuality.
- ▶ Headline trade-off: factuality composite 0.3324 -> 0.4420, ROUGE-Lsum 0.3569 -> 0.3398
- ▶ Bounded scope: train=128, val_tune=64, val_full=128, test=128, qualitative subset =24

Fixture / XSum load Candidate generation SummaC-style scoring FactCC-style scoring ← Entity support ← Weight search ← Audit + packaging

Method and Main Result

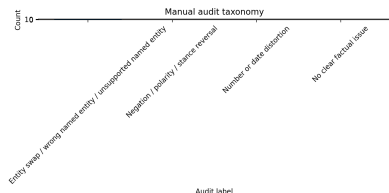
- ▶ Fine-tune facebook/bart-large-xsum on the bounded train split.
- ▶ Generate beam candidates with sizes 4, 8, and 16.
- ▶ Rerank with likelihood, SummaC-style support, FactCC-style consistency, and entity support.
- ▶ Select the operating point on validation search, then evaluate once on the test split.
- ▶ Main result: factuality composite 0.3324 → 0.4420
- ▶ Main trade-off: ROUGE-Lsum 0.3569 → 0.3398
- ▶ Bootstrap result: factuality CI stays positive; ROUGE CI stays slightly negative

Search and Refinement



- ▶ Selected operating point: custom_0097, beam 16
- ▶ Factuality-heavy weights: logprob=0.0, summac=0.75, factcc=1.0, entity=0.5
- ▶ Refinement: add entity support after error analysis
- ▶ Ablation: 0.4061 → 0.4106

Qualitative Analysis and Limits



- ▶ 24-example stratified qualitative analysis with Codex / AI-assisted expert adjudication
- ▶ Outcome counts: reranker win 8, baseline win 8, tie/close 8
- ▶ Dominant remaining failure: entity distortion (12/24)
- ▶ This is a bounded study, not a benchmark-scale XSum claim

Takeaways

- ▶ Simple reranking signals materially improve factuality on this bounded XSum run.
- ▶ Entity support is a useful small refinement, not a silver bullet.
- ▶ The strongest remaining gap is entity-level hallucination when all candidates drift the same way.
- ▶ The next strongest improvements would likely come from better candidate diversity, stronger entity constraints, and broader external evaluation.